

WEB BASED RAIN GAUGE CALIBRATOR

KALIBRATOR PENGUKUR CURAH HUJAN BERBASIS WEB

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ABSTRACT

Rain gauges in Indonesia need recalibration at least twice a year, viz. at the end of dry season where the rain gauges will soon work harder and at the end of rainy season where usually there are many garbage on each rain gauge orifice. Based on the direct surveys in the fields and discussions at some rain gauge calibration laboratories in Indonesian western, central, and eastern parts, the calibration officers in the visited offices generally used manual rain gauge calibrators to recalibrate rain gauges on their fields. Therefore, the dependency of calibration results to the human beings is very high. Meanwhile to avoid data gathering discontinuation, rain gauges should be recalibrated in situ. To cope with this problem and other issues related to rain gauge calibrators, a research to develop a web based rain gauge calibrator was carried out. The research result using research and development method showed that this calibrator can work well as an in situ and ex situ rain gauge calibrator.

Keywords: rainfall, rain intensity, rain gauge, simulator, calibrator, web

ABSTRAK

Pengukur curah hujan di Indonesia membutuhkan kalibrasi ulang sekurang-kurangnya dua kali setahun, yaitu pada akhir musim kemarau di mana pengukur curah hujan akan segera bekerja lebih keras dan pada akhir musim hujan di mana biasanya terdapat banyak sampah di setiap mulut pengukur curah hujan. Berdasarkan survey langsung di lapangan dan diskusi di beberapa laboratorium kalibrasi pengukur curah hujan di Indonesia bagian barat, tengah, dan timur, petugas-petugas kalibrasi di kantor yang telah dikunjungi pada umumnya mereka menggunakan kalibrator pengukur curah hujan manual pada saat mengkalibrasi ulang

pengukur curah hujan di lapangan. Oleh karena itu, ketergantungan hasil kalibrasi pada manusia sangat tinggi. Sementara itu untuk menghindari penghentian pengumpulan data, pengukur curah hujan seyogyanya dikalibrasi ulang di lapangan. Untuk mengatasi masalah ini dan persoalan-persoalan lainnya yang terkait dengan kalibrator pengukur curah hujan, dilakukan suatu penelitian untuk mengembangkan kalibrator pengukur curah hujan berbasis web. Hasil penelitian dengan menggunakan metode penelitian dan pengembangan lapangan menunjukkan bahwa kalibrator ini dapat bekerja dengan baik sebagai kalibrator pengukur curah hujan di lapangan dan di laboratorium.

Kata Kunci: *curah hujan, intensitas hujan, pengukur curah hujan, simulator, kalibrator, web*

1. INTRODUCTION

Existing rain gauge calibrators utilized at rain gauge stations and laboratories in Indonesia generally have some limitations that need to solve immediately. The field calibrators for rain gauges are still manual (Figure 1), so the calibration results must be typed, processed, and printed by the calibration officers. Moreover, water intensities in the calibrators are not constant. Their maximum water intensity is limited by the local gravity. These limitations especially in water intensity stability make the existing rain gauge calibrators are only suitable to calibrate observatory type rain gauges, but they are not appropriate to calibrate rain gauges such as tipping bucket type. Tipping bucket rain gauges – that able to measure rainfall and rain intensity simultaneously (Wijonarko, 2017) - are known to underestimate the rainfall at higher

intensities (Vasvári, 2005), and are slight overestimation at lower intensities (Shedekar et al, 2009). Hence, the rainfall calibration should be carried out at a constant rain intensity.

An automated dynamic calibration of tipping-bucket rain gauges has been applied more than two decades ago (Humphrey & Istok, 1997). The existing automatic (Figure 2) and more over manual rain gauge laboratory calibrators are still not suitable to recalibrate rain gauges in situ, because these calibrators are relatively heavy, big, and vulnerable to rupture. Meanwhile, rain gauges should be recalibrated in situ to reduce the risk of data gathering losses, rain gauge sensor position shifting, and damages due to transportation.



Figure 1. Existing field rain gauge calibrators



Figure 2. Existing automatic rain gauge calibrators at laboratories in Jakarta and Bandung

It is clear that there was a need to solve the aforementioned problems by developing an automatic rain gauge calibrator that is suitable not only for ex situ, but also for in situ disseminations. Ex situ calibration is

used for initial calibration while in situ calibration is utilized for rain gauge recalibrations. For that sake, the aim of this research was to construct a web based rain gauge calibrator.

2. BASIC THEORY

Rain is one kind of precipitation (WMO, 2008, 2012). Precipitation is the liquid or solid product of condensation or sublimation of water vapor that falls down out of clouds or groups of clouds and reaches the earth's surface, while rainfall is the quantity of rain or the volume of water per unit area (1 m^2) that reaches the earth's surface during the observation period (hour, day, etc.) in liquid form (KNMI, 2000). Although it looks simple, measuring rainfall is one of the most difficult challenges in meteorology because of its extreme spatial and temporal variability (Wang, Fisher, & Wolff, 2007).

Rain intensity is the thickness of a liquid layer (mm) per second (KNMI, 2000). Hence, the rain intensity (mm/s) that represents the rate of water input (Jackson, 1975) is the rain fall per second. According to Nemec (1969), measurement errors of tipping bucket rain gauges can be significant during heavy rainfall or light drizzle (Humphrey & Istok, 1997). Hence, rainfall measurements should be related to their intensities, especially at high rainfall intensity (Lelièvre, 2014). The rainfall intensity is also closely linked with raindrop size-distribution (Brandt, 1989) and drop volume (Herwitz, 1995; Calder, 1996a, 1996b; Hall, 1992, 2003; Luo et al, 2013).

This factors can be avoided using constant flows for rain gauge calibrators.

Rain gauge is an instrument to obtain the ground truth in rainfall monitoring (Lanza & Stagi, 2002; Lanza et al. 2007; Lanza, Vuerich, & Gnecco, 2010;). Globally, all types of rain gauges can be divided into catching and non-catching instruments (Lanza & Stagi, 2002; Lanza, 2005; Lanza et al., 2007; Lanza & Stagi, 2008; Lanza & Stagi, 2009; Lanza, Vuerich, & Gnecco, 2010; Lanza, Stagnaro, & Cauteruccio, 2018). Although there are some kinds of catching rain gauges such as double decker tipping bucket (Wijonarko, 2017), the most favorite types are observatory, Hellmann, and tipping bucket (Maftukhah, Wijonarko, & Rustandi, 2016). Even, remote-sensing measurements (satellite, radar, etc) can not substitute ground-based observations (Plummer, Allsopp, & Lopez, 2003). Remote-sensing measurements use non-catching instruments while ground-based observations utilize catching instruments. Besides its main function, the rain gauge can be utilized to measure some other variables such as interception (Wijonarko & Maftukhah, 2014), rain rose (Wijonarko, 2018; Prakosa, Wijonarko, & Rustandi, 2018) and water balance (Wijonarko & Maftukhah, 2016).



Figure 3. A rain simulator with its controller at Sabo Laboratory, Yogyakarta, Indonesia

Rainfall simulator is an equipment to imitate rainfall characteristics such as rainfall duration and quantity to predefined values (Wildhaber et al, 2012). According to Parsakhoo et al (2012) and Corona et al (2013) criteria, the rain simulator in this web rain gauge calibrator is categorized as a nozzle rainfall simulator. Although there are several problems related to the reproducibility of natural rainstorms (Corona et al., 2013), a simulator is very useful equipment to replicate rainfall for the testing other variables such as infiltration (Aksoy et al., 2012), interception (Putuhena & Cordery, 1996; Pypker et al., 2005), erosion (Jomaa et al, 2013), runoff (Zeng et al., 2018) and water management in general (Jackson, Reichard, & Connell, 2018)

Calibrator is the main equipment to conduct a calibration activity. Calibration is a process to compare an instrument under test, such as a rain gauge, with its standard.

Calibration is a method to solve the difference between a measurement and its model (Beven, 1989) and is a part of activity to quality control of measurements (Begeš et al., 2015). A tipping bucket calibration is usually carried out by giving a known quantity of water via the tipping mechanism at various intensities and by adjusting the mechanism to the known volume (WMO, 2008, 2012).

Web or more accurately website is a collection of information in the form of text, picture, video, sound and any other formats that are blended and saved together in a server hosting. The ability of a website to store information that can be downloaded at every internet accessible place is used in this research to build a web based rain gauge calibrator.

3. METHOD

This research used the Research and Development method. There were some activities for this method, namely the problem formulation, design, construction, testing, and development.

The problem formulation in this research was obtained from the discussion result between the representatives of Research Center for Metrology – Indonesian Institute of Sciences and Center for Calibration, Instrumentation, and Engineering – Indonesian Meteorology, Climatology and Geophysics Agency in August 20, 2015, in Jakarta (Figure 4). The problem formulation was strengthened by an informal meeting between the representatives of Research Center for Metrology – Indonesian Institute of Sciences

and Calibration Department, Minister for Public Works and Human Settlements.

The design was conducted as soon as the problem essence has been formulated. This desk study was arranged based on the units. The design result then was used as the main tool to decide components.

The manufacture was utilized to realize every units of the web based rain gauge calibrator. The hardware making was only for the calibration unit (Figure 5), while software development was conducted for all units (calibration unit, data processing unit, and calibration authentication unit).

The testing was conducted to every unit (subsystem testing) and the integration for all units (system testing). The system testing was conducted in the laboratory and in the field (Figure 6).

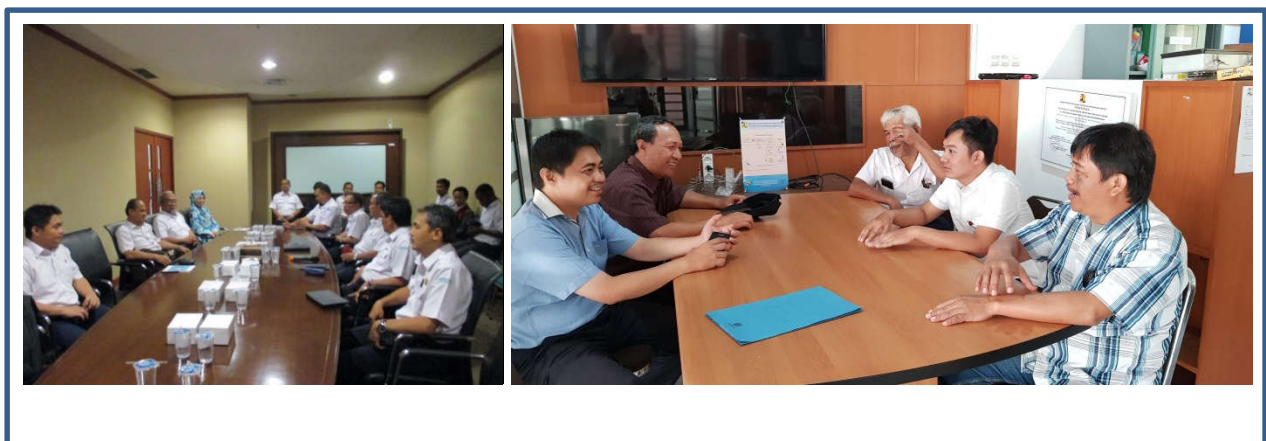


Figure 4. Discussions as the way to get the problem formulation for this research in Jakarta and Bandung

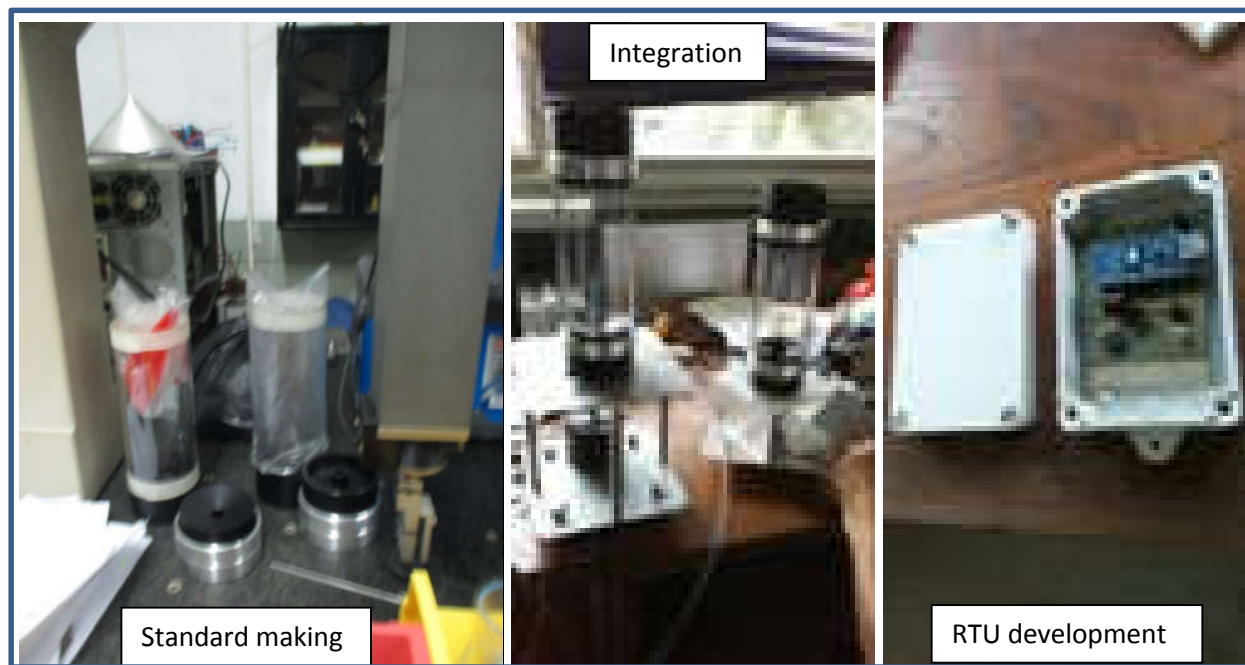


Figure 5. Calibration unit making process



Figure 6. Calibration testing in the laboratory and in the field

The testing result was used as the input for the development. If there was no flaw, damage, or improvement idea, no further development was needed.

4. RESULT AND DISCUSSION

This research has produced a web based rain gauge calibrator. The web rain gauge calibrator is an instrument to calibrate rain

gauges *ex situ* or *in situ*, but calibration results can be downloaded from the web developed for this necessity by the technical manager and authorized personnel on his desk. Besides functioning as a working calibrator - a standard that is used routinely to calibrate or check material measures, measuring instruments or reference materials (WMO 2008, 2012) - this calibrator also works as a rain simulator in a very simple construction form to the rain gauge under test.

The web based rain gauge calibrator patented at P00201707655 consists of three integrated units (Wijonarko *et al.*, 2017), namely calibration unit, data processing unit, and calibration authentication unit. The first two units are mobile units while the last unit is any computer that can communicate with the internet, but formally is located at the technical manager office.

The calibration unit comprises a standard part and a data logger (Figure 7) which is superimposed on the mouth of rain gauge under test. Santana, Guimarães, & Lanza (2018) call this rain gauge under test is calibrated as a stand-alone instrument.

The core of standard part is a 997 ml acrylic cylinder - calibrated in 2017 to National Metrology Institute in Indonesia (Figure 11) - that is used to simulate rainfall to the rain gauge under test. For rain gauge calibration, the cylinder is filled with water from a tap or dipper through a custom made funnel until the water reaches the cylinder top. The water then spills through a plastic hose to the ground or any water dump. The funnel is used to eliminate air bubbles trapped and attached to the inner part of cylinder wall. The water valve at the bottom side of cylinder is then opened based on a software instruction. The water flows at a certain water intensity to the mouth of rain gauge under test. The rain gauge under test responds water enters to its input by giving a pulse output. Every pulse is equal to a certain water volume. This volume depends on the resolution and mouth area of the rain gauge sensor. The output pulse is connected via one way serial cable to the data logger. The data logger then counts, saves, and sends the data to the data processing and calibration result presentation unit via a USB to USB cable.

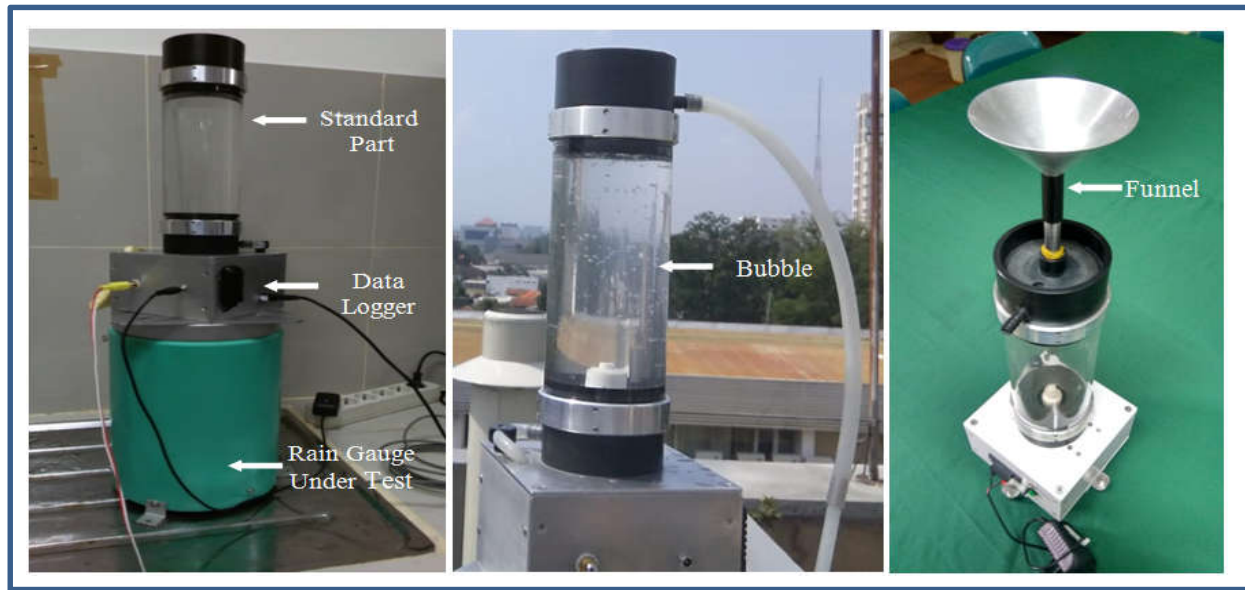


Figure 7. Two calibration units above rain gauges under test in the laboratory and in the field respectively

The data processing unit is a lap top and the like with application software (Figure 8) that conducts some steps. First, the data processing unit is connected to the calibration unit. Second, this unit is filled with the specification of the rain gauge sensor under test especially the resolution, mouth diameter or area, brand, type/serial number, and order number; and calibration operator, time, location, and rainfall intensity. The percent of absolute bias is

largest for small rainfall intensities (Muñoz, Céleri, & Feyen, 2016). The sensor orifice diameter checking is one of rain gauge calibration procedure (WMO, 2008, 2012). Third, the program is executed. This program will initialize a calibration process, count the output of rain gauge under test, measure ambient variables, process the data, display the result and send information to its web.

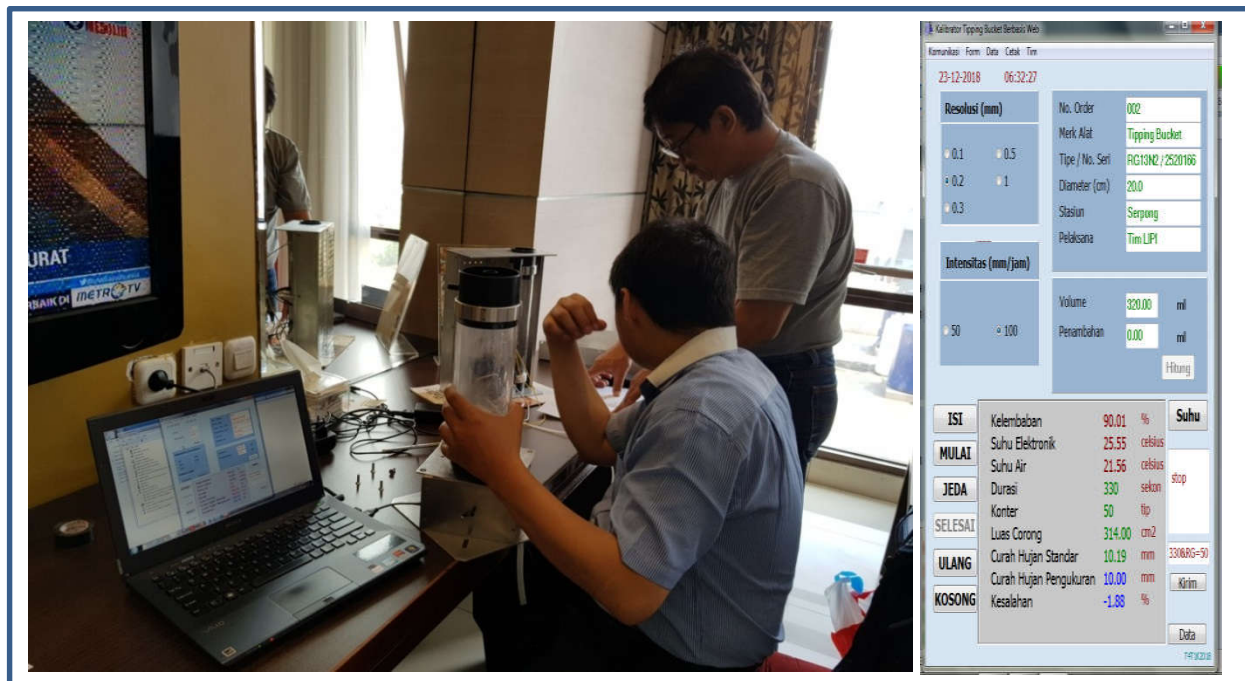


Figure 8. The hardware, software and brainware for data processing unit

The calibration authentication unit is basically a printer connected to a computer (Figure 9). This computer can be any computer connected to the internet network to retrieve information sent by the data

processing unit. The information is printed as a calibration certificate. This certificate is signed by the the Technical Manager for calibration.

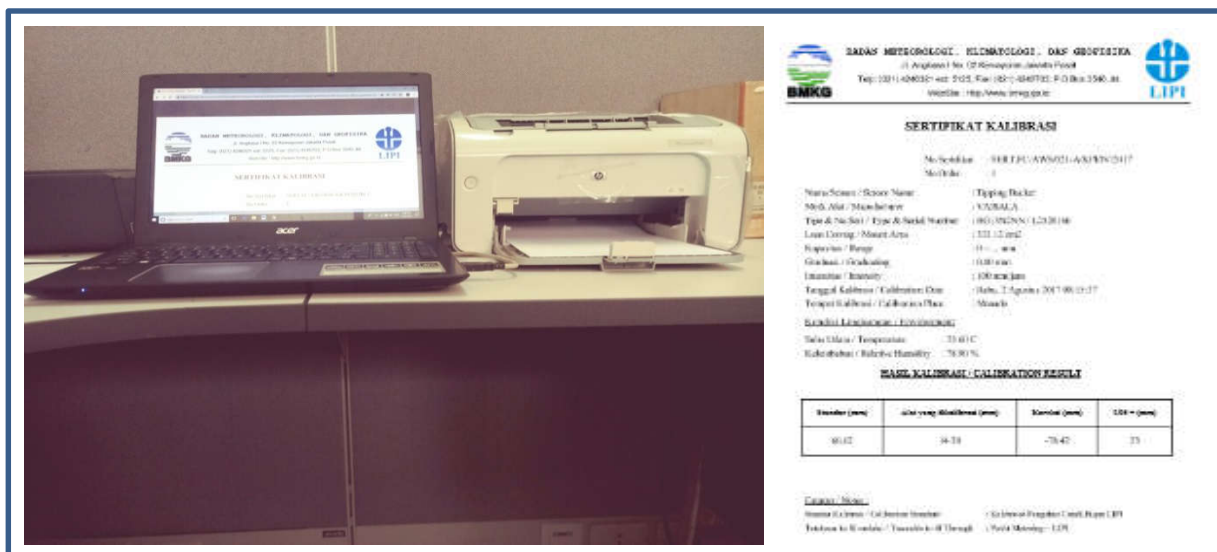


Figure 9. The calibration authentication unit and its calibration certificate

The web based rain gauge calibrator is especially used to calibrate automatic rain gauges such as tipping bucket type. This selection is based from the result of the first Laboratory Intercomparison of Rainfall Intensity Gauges performed by WMO from September 2004 to September 2005 that shows that only those tipping-bucket rain gauges that apply proper correction to account for mechanical errors comply with the WMO specifications on the required accuracy for rainfall intensity measurements (Lanza, 2005).

For manual type such as observatory and Hellmann types, the cylinder volume is compare to the burette of the manual rain gauges (Figure 10). Reliability of calibration is in fact strictly connected with the

capability in controlling and managing inherent calibration uncertainties (Molini et al, 2005). The web based rain gauge calibrator itself was calibrated to the Indonesian NMI. The uncertainty – information that tells us how good this measurement is (Beamex, 2016, 2017) - of this calibrator was around 0.3 % (Figure 11). The uncertainty of rain gauges usually is around 5 %. Therefore, this calibrator is suitable for calibrating rain gauges. Calibrating rain gauges will improve the quality rainfall measurements. The accuracy rainfall data determines the accuracy of other variables such as water balance (Muñoz, Tume, & Ortiz, 2014; Wijonarko & Maftukhah, 2016).



Figure 10. Manual rain gauges and their burette



Figure 11. The calibration certificate for this web based rain gauge calibrator

From the aforementioned discussion, it can be seen that this web based rain gauge calibrator has many technological innovations that have not been found yet in the previous calibrators. The web based rain gauge calibrator provides laboratory class calibration data, although the calibration is carried out in situ. This calibrator reduces human error and presents results directly to the technical manager as long as there is a good internet network. Thus, this calibrator helps us to ensure the quality of rainfall data recorded by the rain gauge under test.

From epistemological aspect, this research gives a significant result. Initially rain gauge calibration method used static calibration method. The static method was improved to dynamic calibration by Calder & Kidd (1978). The dynamic calibration

method was further improved to automated dynamic method by Humphrey & Istok (1997). The automated dynamic method is revised by this research to web based calibration method.

Ideally, every installed rain gauge in Indonesia needs recalibration at least twice every year, viz. at the end of dry season where the rain gauges will soon work more often and the end of rainy season where garbage might clog the rainfall flow to the rain gauge. The rainfall ratio between season (for example between winter and summer) depends on the spatial aspect (Akhtar, Ahmad, & Booi, 2008). Hence, the right time to recalibrate every rain gauge should be based on each indigenous rainfall characteristic.

Rain gauge diameters have many sizes. There is a possibility that the volume

of this web based rain gauge calibrator is not right fitted with the needed size. The build-up and build-down method (Sardjono & Wijonarko, 2018) might be used to solve this problem.

The accuracy of rainfall radar is assessed by rain gauges (Sebastianelli et al, 2013; Jackson, Reichard, & Connell, 2018). Hence, this web based rain gauge calibrator should not only limited to calibrate contact rain gauges, but also for non-contact rain gauges over land.

This is caused by the fact that non-contact rain gauges such as radar provide measurement over a much wider area, but the quantitative measurements are less accurate than those obtained from rain gauges (Luczak, 2002). For small or inaccessible land surface, however, special techniques are needed (Houze Jr. et al, 2004).

5. CONCLUSION AND SUGGESTIONS

Based on the above analysis, it can be concluded that the developed web based rain gauge calibrator worked well to calibrate a rain gauge in situ or ex situ circumstances. Hence, the web based rain gauge calibrator is a real solution for the existing rain gauge calibrators especially the calibrators used in the field.

It is suggested that the web based rain gauge calibrator should be refined with finishing touches by a

hydroclimatological industry to attract consumers. The products then can be bought from the industry and applied by the users to calibrate and recalibrate thousands of rain gauges in the field.

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